

# Athrav Seruwam

647-960-0095 | [athravmk@gmail.com](mailto:athravmk@gmail.com) | [linkedin.com/in/a-seruwam](https://linkedin.com/in/a-seruwam) | [github.com/athravseruwam07](https://github.com/athravseruwam07) | [athravseruwam.com](https://athravseruwam.com)

## EDUCATION

### University of Waterloo

Candidate for BAsC, Mechatronics Engineering

Waterloo, ON

Sept. 2025 – Apr. 2030 (Expected)

## TECHNICAL SKILLS

**Languages:** Python, C/C++, Java, JavaScript, TypeScript, Ruby, SQL

**Frameworks & Tools:** React, Next.js, FastAPI, Node.js, Docker, PostgreSQL, REST APIs

**Cloud & Platforms:** AWS (ECS, CloudFront, RDS, S3, Bedrock), Stripe, YOLO

## EXPERIENCE

### Software Engineer Intern

Jan. 2026 – Apr. 2026

Quotograph

Waterloo, ON

Next.js · Javascript · Python · FastAPI · PostgreSQL · AWS · Stripe

- Engineered permit analysis **AI** features – approval status prediction, PDF export, and admin user management – for a prospective enterprise client; delivery contributed to a **\$200K/year** contract win
- As **sole developer** across 3 commercial SaaS products, architected an **AI Safety Forms** system end-to-end – AI form generation, checklist enforcement, inspection workflows, and QR-tracked equipment audits – shipped to production
- Led **AWS** production deployment of PermitX across **ECS, CloudFront, RDS, S3, and Bedrock**, migrating a 3-service SaaS platform from development to a production-grade cloud environment
- Engineered a cross-service **Stripe** billing system with subscription tiers, referral attribution, and checkout flows; built multi-step document approval with configurable branding and **PDF generation**

### Firmware Developer

Sept. 2025 – Jan. 2026

University of Waterloo Formula Electric

Waterloo, ON

- Engineered embedded **C/C++** firmware for ECUs managing **CAN/LIN** vehicle communication, integrating multi-sensor interfaces across vehicle subsystems
- Built a signal simulation and **HIL testing** framework for control board validation, cutting per-board validation time by **35%** and enabling automated fault detection across overcurrent and bus timeout conditions

### Mechanical Designer Co-op

Oct. 2024 – Jan. 2025

Automha Americas

Oakville, ON

- Designed semi-automated warehouse layouts in **AutoCAD**, integrating racking and conveyance systems to improve storage utilization by **20%** and reduce estimated material handling time by **15%**
- Built **Excel** capacity models to quantify racking throughput, aisle clearances, and storage capacity, informing layout decisions across client projects

## PROJECTS

### Dextera | ESP32, Next.js, TypeScript, WebSockets, Node.js | ConHacks – 2nd Place

- Built a smart-glove rehabilitation platform for stroke and hand-injury recovery; an **ESP32** reads **5 flex sensors** and streams per-finger bend data over **WebSockets** to a Next.js clinician dashboard with live and historical exercise tracking
- Designed session tracking with per-session rep counts, accuracy scores, and flex-range trends to support remote therapist monitoring and week-over-week recovery analysis

### Clarus | Next.js, TypeScript, Tailwind CSS, Playwright, LLMs

- Built an AI academic copilot that scrapes **D2L/Brightspace** via a **Playwright** connector and uses an **LLM** to generate a deadline-ranked daily action plan with per-course study briefs, eliminating manual schedule aggregation
- Engineered the full-stack ingestion pipeline in **Next.js** and **TypeScript**; scraper parses syllabi, due dates, and grade weights and feeds structured data to the LLM context window for personalized task prioritization

### Doceo | Next.js, FastAPI, Gemini, TTS, Canvas API

- Built an AI STEM tutor generating full animated whiteboard lessons with synchronized **Gemini** TTS narration in **10–30 seconds**; implemented a stateful interrupt handler allowing students to redirect lesson context mid-stream without restarting generation
- Architected a **Next.js/FastAPI** pipeline with a custom **Canvas** animation engine and chunked TTS streaming synchronized to real-time **LaTeX** equation rendering